Lab experiment-29

Experiment 29. Backward Chaining

Aim

To implement **Backward Chaining using Prolog**.

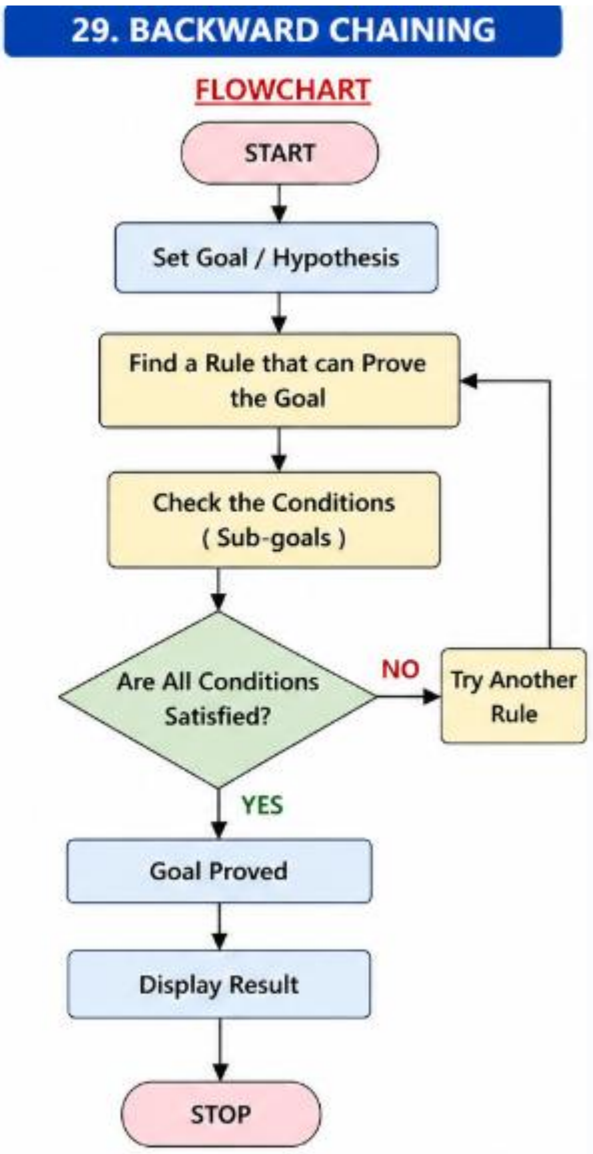
Objectives

- To understand backward reasoning.
- To prove a goal using facts and rules.
- To perform logical inference.

Algorithm

1. Define facts and rules.
2. Set a goal.
3. Find a rule that can prove the goal.
4. Check its conditions.
5. Prove the goal if conditions are satisfied.
6. Display the result.

Flowchart



Code

```
bird(parrot).

has_wings(parrot).

can_fly(X) :-

    bird(X),
```

```
    has_wings(X).

main :-

    Goal = can_fly(parrot),

    call(Goal),

    write('Goal: '),
write(Goal), nl,

    write('Result: Goal
Proved'), nl.

:- initialization(main).
```

Output

Result

Goal: can_fly(parrot)

Result: Goal Proved

Conclusion

Thus, **Backward Chaining was successfully implemented using Prolog**.